

Day 3 : RAM and Storage Theory

Task 1 : Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Typical Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Voltage	~1.5 V	~1.2 V	~1.1 V
Performance	Basic/older	Good	Highest
Typical Use	Older PCs and laptops	Many mainstream PCs and laptops	New-generation gaming PCs, workstations, and high-performance PCs
Power Efficiency	Lowest	Better than DDR3	Best
Current Relevance	Mostly outdated	Still widely used	Modern standard

In simple words: DDR3 is the oldest and slowest of the three, DDR4 offers a good balance of speed and power consumption, while DDR5 provides the highest speeds and is mainly used in newer computers.

Task 2 : Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Practical: RAM Removal and Reinstallation

- I switched off the computer and disconnected the power supply before opening the system.
- I located the RAM module on the motherboard and took a clear photograph of it. After removing and reinstalling the RAM module properly in its slot, I powered on the computer.
- The system booted successfully, and the installed RAM was detected correctly.

Result: RAM was successfully removed and reinstalled, and the computer started normally.

Task 3 : List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD	SSD (SATA)	NVMe SSD
Speed	Slowest	Faster than HDD	Fastest
Technology	Uses spinning magnetic disks	Uses flash memory	Uses flash memory with PCIe interface
Typical Use	Large file storage and backups	Everyday computing and applications	Gaming, heavy software, and fast data processing
Daily App / Use Case	Google Drive files / backups – useful for storing large amounts of files	Spotify – faster app loading and smooth everyday use	PUBG / Free Fire – helps games load much faster and reduces loading times

Three Key Differences

1. **Speed:** HDD is the slowest, SATA SSD is faster, and NVMe SSD is generally the fastest.
2. **Technology:** HDD has moving mechanical parts, while SSD and NVMe use flash memory with no moving parts.
3. **Best Use:** HDD is best for **large and inexpensive storage**, SSD is good for **everyday apps**, and NVMe is best for **high-performance tasks such as gaming and heavy applications**.

Task 4 : Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly

Practical: Storage Drive Removal and Reinstallation

- I switched off the computer and disconnected the power supply before opening the system.
- I located the storage drive and identified its connector as SATA/M.2. I carefully removed the storage drive, took a clear photo of the connector, and then reinstalled the drive properly.
- After reconnecting the power and starting the computer, the system successfully detected the storage drive.

Result: The storage drive was successfully removed and reinstalled, and the device recognized it correctly after reassembly.